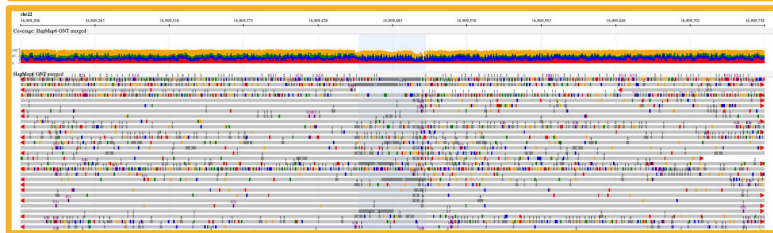
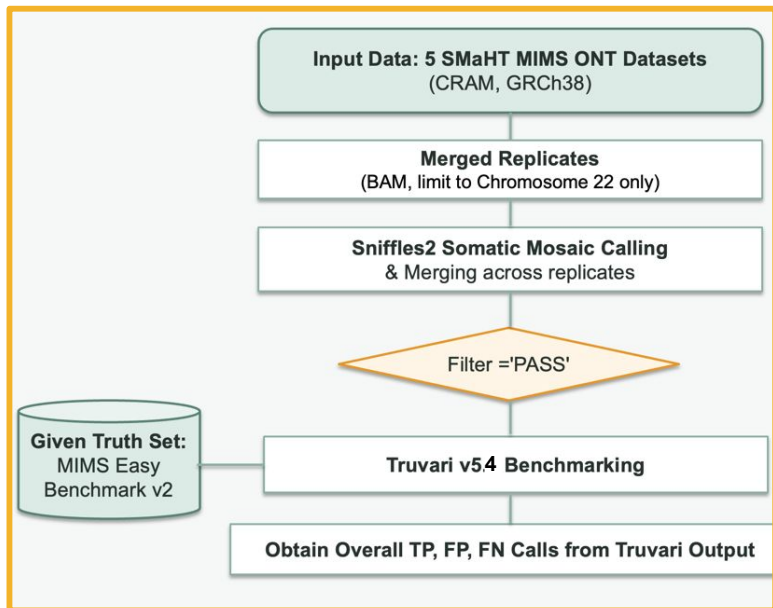


Methods 1: create pilot data of false positive SV calls from Sniffles to manually review, characterize, and inform caller improvements



Datasets and Steps:

- HapMap (“truth set”) + SmaHT data → Sniffles
 - Subset to false positives in chr22
 - Easy vs hard dataset
 - Acts as pilot dataset to review metrics and common IGV findings
- Truvari benchmark (pass-only, chromosome 22)
 - Produced 483 TP, 198 FN, **147 FP** count for manual inspection
 - 147 FP -> 32 Germline 115 Mosaic

Other details:

- Integration of TRF (tandem repeat) and/or mobile element tracks for annotating challenging regions

Visualise:

- Bamsnap-LRS generated IGV plots with all mismatches and SV region annotated

[illegible]

1. **Consider VCF fields for key metrics:**

- ## 2. Manually review variants in IGV and curate characteristics of false calls

- ### 3. Aggregate curated information to inform implementation of altered or new filters for Sniffles

Methods 3: use curated findings to implement caller filter updates and improvements and test impact on SV calls in other chromosomes



Characterize FP Features

- Variant Allele Fraction (VAF)
- Supporting Read Count
- Breakpoint Orientation
- SV Type & Genomic Context
- Tandem Repeats & Chimeric/Foldback Reads

IGV Manual Read-Level Inspection

- Support, Breakpoints, Soft Clipping, Alignment, Context

Analyze Patterns & Develop Filtering/Flagging Criteria

Applying Proposed Flags/Filters & Re-benchmarking (Whole Genome or Targeted Region)

**Final Evaluation: Compare Filtered Sniffles2 Calls
with Given Truth Set**

**Implement and iterate on Sniffles caller
updates**



**Characterize the impact of updates on
false positive calls across other
chromosomes (hopefully less)**



**Characterize the impact of updates on true
positive calls across other chromosomes
(hopefully none)**



**Keep going until time runs out or we have
hit a good output**

